For simplicity, we use a vehicle kinematic model, where we assume that there is no tireslip.

Our state is $z(t) = \begin{bmatrix} p_x(t) \\ p_y(t) \\ \theta(t) \\ v(t) \end{bmatrix}$ our control input is $u(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix}$ where $a(t)$: longitudinal acceleration, $\delta(t)$: steering angle.

where p_x, p_y is the position of the rear-axle midpoint in a fixed ground coordinate system

θ heading of car (in radians)

v Forward speed (metres/second)

For the rear-axle reference point:

$$\dot{p}_x = v\cos\theta \qquad \dot{p}_y = v\sin\theta$$

Steering radius is: $R = \dfrac{L}{\tan\delta}$

$$\Rightarrow \quad \dot{\theta} = \frac{v}{R} = \frac{v}{L}\tan\delta$$

$$\Rightarrow \quad \dot{v} = a$$

$$\Rightarrow \quad \dot{z} = \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{\theta} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ \frac{v}{L}\tan\delta \\ a \end{bmatrix} = f(z,u)$$

Next, we need to derive the error-based deviation model:

$$e(t) = \begin{bmatrix} e_x(t) \\ e_y(t) \\ e_\theta(t) \\ e_v(t) \end{bmatrix}$$

where $e_x = p_x - p_{x,\text{ref}}$; $e_y = p_y - p_{y,\text{ref}}$; $e_\theta = \text{wrap}_{[-\pi,\pi)}(\theta - \theta_{\text{ref}})$;

$$e_v = v - v_{\text{ref}}$$

$$\Rightarrow \quad \dot{e}_x = v\cos\theta - \dot{p}_{x,\text{ref}} \quad ; \quad \dot{e}_y = v\sin\theta - \dot{p}_{y,\text{ref}}$$

- How far can the vehicle deviate from its reference over the next τ seconds?

$$\text{Let } p(t) = \begin{bmatrix} p_x(t) \\ p_y(t) \end{bmatrix}, \qquad e_p(t) = p(t) - p_{\text{ref}}(t)$$

$$\|e_p(t)\|_2 = \sqrt{e_x(t)^2 + e_y(t)^2}$$

We assume the following:

$$\|\dot{e}_p(s)\|_2 \le \bar{q}$$

$$\|\dot{p}(s) - \dot{p}_{\text{ref}}(s)\|_2 \le \bar{q} \qquad \forall\, s \in [t, t+\tau]$$

where $\bar{q}$ is velocity-vector mismatch, which measures differences in speed and direction

Here we are asking:

Knowing that e_p cannot change faster than $\bar{q}$, how large could it become after τ seconds?

Derive the future bound error:

since

$$\dot{e}_p = \dot{p} - \dot{p}_{\text{ref}}$$

$$\Rightarrow \quad e_p(t+\tau) = e_p(t) + \int_t^{t+\tau} \dot{p}(s) - \dot{p}_{\text{ref}}(s)\, ds$$

Triangular inequality:

$$\|e_p(t+\tau)\|_2 \leq \|e_p(t)\|_2 + \int_t^{t+\tau} \|\dot{p}(s) - \dot{p}_{\text{ref}}(s)\|_2\, ds$$

$$\|e_p(t+\tau)\|_2 \leq \underbrace{\|e_p(t)\|_2 + \bar{q}\tau}_{\rho(\tau)}$$

$\Rightarrow \rho(\tau)$ is a conservative radius around the future reference position.

It represents a circle within which we think our vehicle will be with respect to a reference point in the future.

We need to derive $\bar{q}$:

Let h be the unit vector pointing along the heading θ

$$h(\theta) = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$\Rightarrow \quad \dot{p} = v\, h(\theta), \qquad \dot{p}_{\text{ref}} = v_{\text{ref}}\, h(\theta_{\text{ref}})$$

$$\Rightarrow \quad \dot{p} - \dot{p}_{\text{ref}} = v h(\theta) - v_{\text{ref}}\, h(\theta_{\text{ref}})$$

We want to separate the sources of mismatch, so add and subtract $v_{\text{ref}}\, h(\theta)$

$$\Rightarrow \quad \dot{p} - \dot{p}_{\text{ref}} = \overbrace{(v - v_{\text{ref}})h_\theta}^{\text{speed mismatch}} + \overbrace{v_{\text{ref}}\big(h(\theta) - h(\theta_{\text{ref}})\big)}^{\text{direction mismatch}}$$

Apply triangle inequality:

$$\|\dot{p} - \dot{p}_{\text{ref}}\| \leq |e_v| + |v_{\text{ref}}|\, \|h(\theta) - h(\theta_{\text{ref}})\|_2$$

Two unit vectors separated by angle e_θ have distance

$$\|h(\theta) - h(\theta_{\text{ref}})\|_2 = 2\left|\sin\left(\frac{e_\theta}{2}\right)\right|$$

$$\Rightarrow \quad \|\dot{p} - \dot{p}_{\text{ref}}\|_2 \leq |e_v| + 2\,|v_{\text{ref}}|\left|\sin\left(\frac{e_\theta}{2}\right)\right|$$

if we can establish a bound throughout that interval:

$$|e_v| \leq \bar{e}_v \ ; \ |e_\theta| \leq \bar{e}_\theta \leq \pi, \quad v_{\text{ref}} \leq \bar{v}_{\text{ref}}$$

$\Rightarrow$ we can calculate:

$$\boxed{\bar{q} = \bar{e}_v + 2\,\bar{v}_{\text{ref}} \sin\left(\frac{\bar{e}_\theta}{2}\right)}$$

We can try to derive those bounds experimentally, where we could potentially have a look up table or something.

- How much space is available around the reference trajectory before the vehicle touches an object?

We enclose the vehicle within a disk of radius r_{veh} and center the rear axle midpoint

At future time $t + \tau$, the reference point can be displaced from its desired position by at most:

$$\rho(\tau) = \rho_0 + \bar{q}\tau$$

$\Rightarrow$ the entire vehicle is contained within a disk centered at $p_{\text{ref}}(t + \tau)$ with radius

$$r_{\text{veh}} + \rho_0 + \bar{q}\tau$$

Let $\mathcal{O}_k$ be the region occupied by obstacle k:

$$\Rightarrow \quad d_k(\tau) = \text{dist}\left(p_{\text{ref}}(t+\tau), \mathcal{O}_k\right)$$

This is the shortest distance from the future reference point to any point in the obstacle, rather to its center.

$$\Rightarrow \quad d_k(\tau) = \max\left(\|p_{\text{ref}}(t+\tau) - o_k\|_2 - r_k,\ 0\right)$$

where o_k is center of $\mathcal{O}_k$ and r_k is the radius

$\Rightarrow$ a sufficient condition for avoiding obstacle k at the future time is:

$$d_k(\tau) > r_{\text{veh}} + \rho_0 + \bar{q}\tau$$

$\Rightarrow$ clearance margin is:

$$m_k(\tau) = d_k(\tau) - r_{\text{veh}} - \rho_0 - \bar{q}\tau$$

if $m_k(\tau) > 0 \Rightarrow$ we are clear from objects
if $m_k(\tau) \leq 0 \Rightarrow$ we are no longer safe, a collision is possible within the bound.

If we have a dynamic object:

We want to construct a region containing every possible future position of one moving object

At time t, let:

- $\hat{o}_k$: estimated position of object k
- $\hat{w}_k$: estimated velocity vector
- $\varepsilon_{p,k}$: bound on position-estimation error
- $\varepsilon_{v,k}$: bound on velocity-estimation error
- $\bar{a}_k$: bound on the object's acceleration magnitude throughout the prediction interval.

$$\Rightarrow \qquad \|o_k(t) - \hat{o}_k(t)\| \leq \varepsilon_{p,k}$$
$$\|w_k(t) - \hat{w}_k(t)\| \leq \varepsilon_{v,k} \qquad \|a_k(t+s)\| \leq \bar{a}_k$$

These are assumptions we must experimentally validate.

In our settings, we can pass these values through simulator + injection of noise, however we need to find the different current benchmarks we have for such tasks to simulate accordingly.

if the object maintained its estimated velocity, its position after τ seconds would be:

$$\hat{o}_k(\tau) = \hat{o}_k + \hat{w}_k \tau$$

This gives the center of our future region. We do not assume the object actually maintains constant velocity.

$$\Rightarrow \quad o_k(t+\tau) = o_k(t) + w_k(t)\tau + \int_0^\tau (\tau - s)\, a_k(t+s)\, ds$$

$$\|o_k(t+\tau) - \hat{o}_k(\tau)\|_2 \leq \varepsilon_{p,k} + \varepsilon_{v,k}\tau + \bar{a}_k \int_0^\tau (\tau - s)\, ds$$

$$\Rightarrow \quad \boxed{\beta_k(\tau) = \varepsilon_{p,k} + \varepsilon_{v,k}\tau + \frac{1}{2}\bar{a}_k \tau^2}$$

We just found $\beta_k(\tau)$ which is the position uncertainity region

$$\Rightarrow \quad R_{\text{occupancy},k}(\tau) = r_k + \beta_k(\tau) \quad \text{with } r_k \text{ the radius of the object.}$$

$\Rightarrow$ No potential collision if circles are disjoint:

$$\|p_{\text{ref}}(t+\tau) - \hat{o}_k(\tau)\|_2 > r_{\text{veh}} + r_k + \rho_0 + \bar{q}\tau + \varepsilon_{p,k} + \varepsilon_{v,k}\tau + \frac{1}{2}\bar{a}_k \tau^2$$

$\Rightarrow$ Define the predicted center distance:

$$D_k(\tau) = \|p_{\text{ref}}(t+\tau) - \hat{o}_k(\tau)\|_2$$

Define required seperation

$$S_k(\tau) = r_{\text{veh}} + r_k + \rho_0 + \varepsilon_{p,k} + (\bar{q} + \varepsilon_{v,k})\tau + \frac{1}{2}\bar{a}\tau^2$$

$$\Rightarrow \quad m_k(\tau) = D_k(\tau) - S_k(\tau)$$

Therefore $m_k(\tau) > 0$: the enclosing disks are seperated
$m_k(\tau) < 0$: our bounds no longer certify clearance

$\Rightarrow$ The condition over all obstacles will be:

$$\min_k m_k(\tau) > 0 \qquad \forall\, \tau \in [0; H]$$

Let $T_{\text{cap}} = \min(T_{\text{rem}}, T_{\text{valid}})$

where T_{rem} is the remaining reference duration and T_{valid} is how far our model and uncertainty assumptions hold

$$\Rightarrow \quad H_{\text{clear}} = \inf\left\{\tau \in [0, T_{\text{cap}}] : \min_k m_k(\tau) \leq 0\right\}$$

inf means the earliest boundary time